Substances & separation

Identify & represent

1 What is the substance?

Element: atoms of one element.
Compound: different elements chemically combined in a fixed ratio.

Mixture: substances retain chemical properties and can be separated physically. Compounds need chemical reactions.

2 Symbols & equations

H_2O = 2H and 1O; $Ca(OH)_2$ = 1Ca, 2O, 2H. Use correct upper/lower case in symbols.

$2Mg + O_2 \rightarrow 2MgO$. Count each element on both sides; change coefficients only.

Binary names often end -ide; -ate names usually include oxygen.

Separate solids

3 Filtration

Separates **insoluble solid** from liquid. Residue stays on filter; filtrate passes through.

Dissolved salt passes through too. Salt + sand: dissolve salt, filter sand, crystallise filtrate.

4 Crystallisation

Partly evaporate solvent → concentrated solution → cool → crystals form.

Filter crystals and dry them. Avoid heating hydrated salts strongly to complete dryness.

Wear goggles; handle hot glassware carefully.

Separate liquids

5 Distillation

Simple: boil liquid, cool vapour in condenser, collect liquid distillate. Recover water from salt solution.

Fractional: column gives repeated evaporation/condensationto separate miscible liquids by boiling point.

Lower-boiling fraction usually comes first; cooling water enters condenser at bottom.

6 Chromatography

Components have different attractions to solvent and stationary phase, so travel different distances.

Pencil baseline **above solvent;** small spots; mark solvent front. Ink could dissolve; submerged spots wash off.

No new substances form in these physical separations.